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Every camera is a structural health sensor.

SPANDAN Engine · Civic Vibrometry

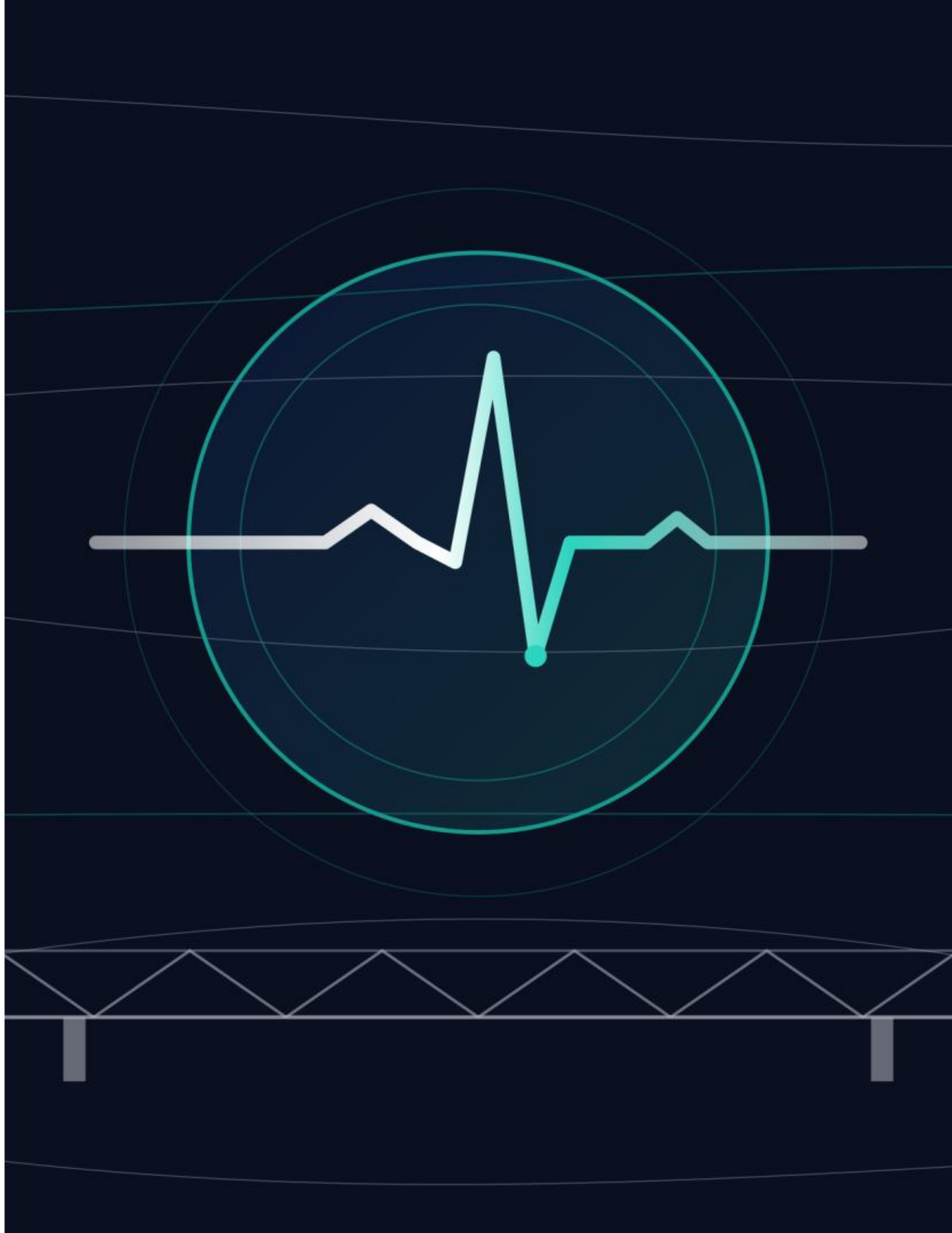
A phone hears a structure's heartbeat from ordinary video — and says when it changed, where the damage is, what caused it, and how sure it is. Non-contact. One operator. 60 seconds.

0.31% frequency error · 0.10% vs contact sensor · 167 Hz from a 60 fps camera

damage located + mechanism named + conformal confidence · DIN 4150-3 / IS-ISO 4866 screening

AI ARENA 3.0 · TEAM_ALPHA · AI VISION

LIVE: tremorlens-live.vercel.app · github.com/mightbeanshuu/tremorlens



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1,72,517

structures inventoried by India's Bridge Management System.

0

with a published record of their structural heartbeat — the fingerprint that shifts when they weaken.

This submission is the instrument that changes the second number.

IBMS rates what inspectors can see. Nothing measures what they cannot.

01 — PROBLEM UNDERSTANDING

India inspects its bridges by eye.

Morbi 2022 — 135 dead. Gambhira 2025 — 22 dead: a crushed pedestal no eye could see. ~170 collapses, 202 deaths, 2021-25.

80% of analysed failures are flood/scour driven — and scour announces itself as a frequency drop no inspector can see.

MoRTH now mandates pre- and post-monsoon inspections (IRC:SP:35). The mandate exists. The instrument did not.

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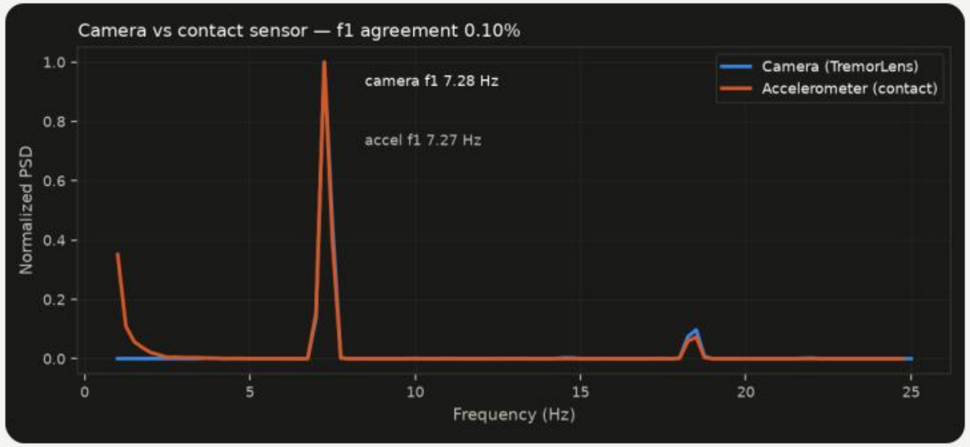
02 — CONCEPT DEVELOPMENT

Structures have a heartbeat. Cameras can hear it.

In 1816 a rolled sheet of paper became the stethoscope, and medicine went from guessing to listening. Civil engineering is still in 1815 — it inspects by looking.

TremorLens reads sub-pixel motion in ordinary video (MIT lineage: Wu 2012, Wadhwa 2013), recovers the structure's natural frequencies, and diagnoses drift against its own healthy baseline. Chosen over 19 scored alternatives.



Camera vs contact accelerometer — same heartbeat, 0.10% apart.

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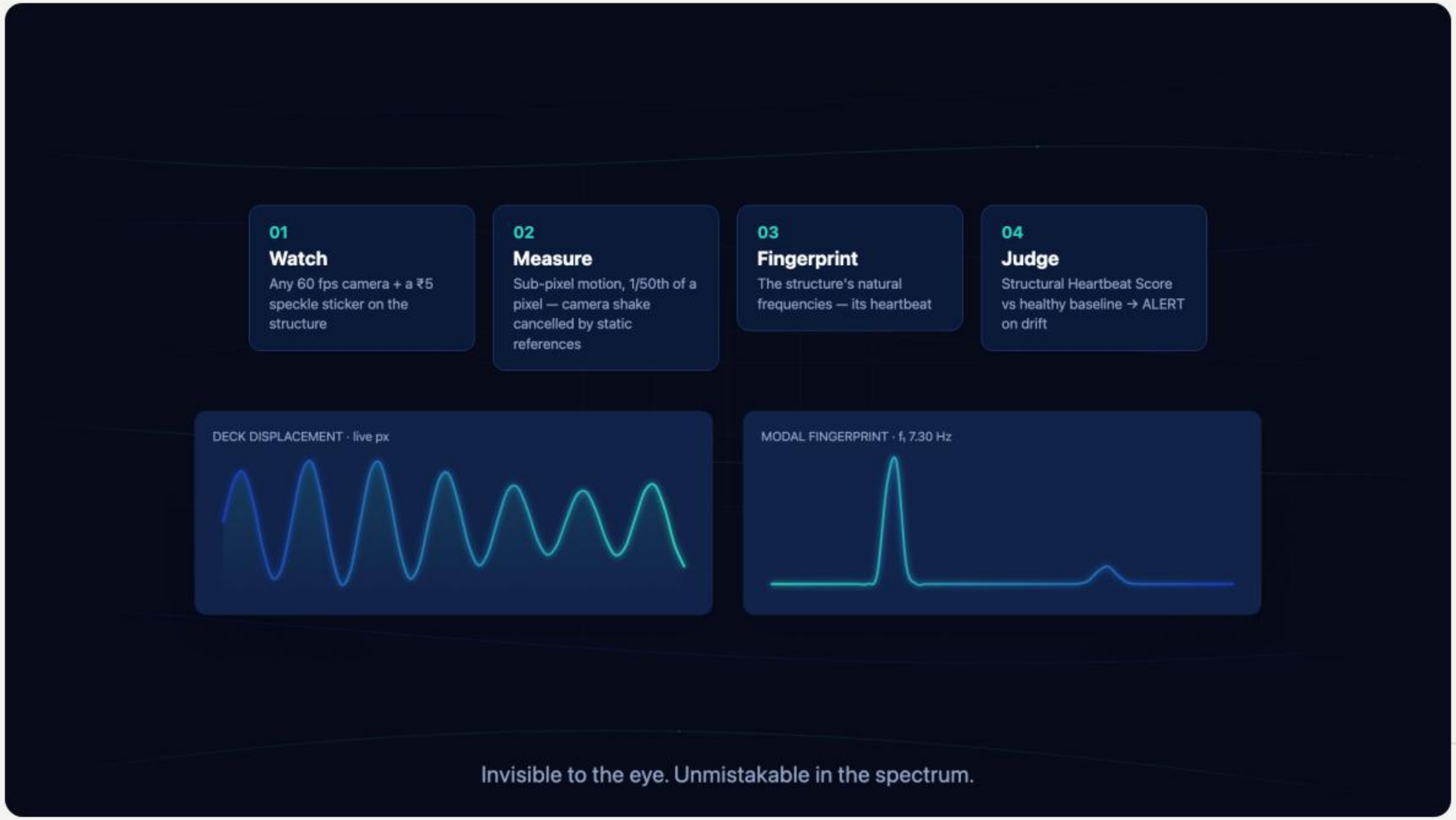
03 — SYSTEM DESIGN — HIGH-LEVEL

One pipeline, camera to verdict.



Watch — any 30-60 fps camera plus a ₹5 speckle sticker.

Measure — ~1/50-px tracking, shake-cancelled by static references.

Fingerprint — frequencies, damping, voice quality.

Judge — SPANDAN: detect, locate, name the mechanism, quantify confidence.

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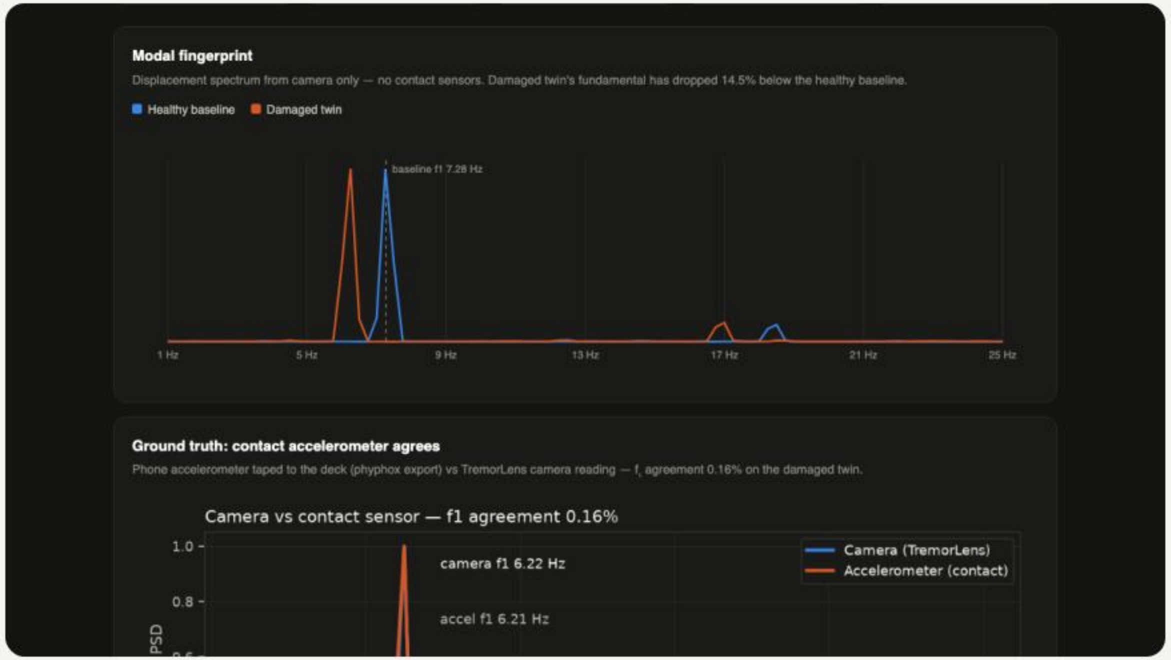
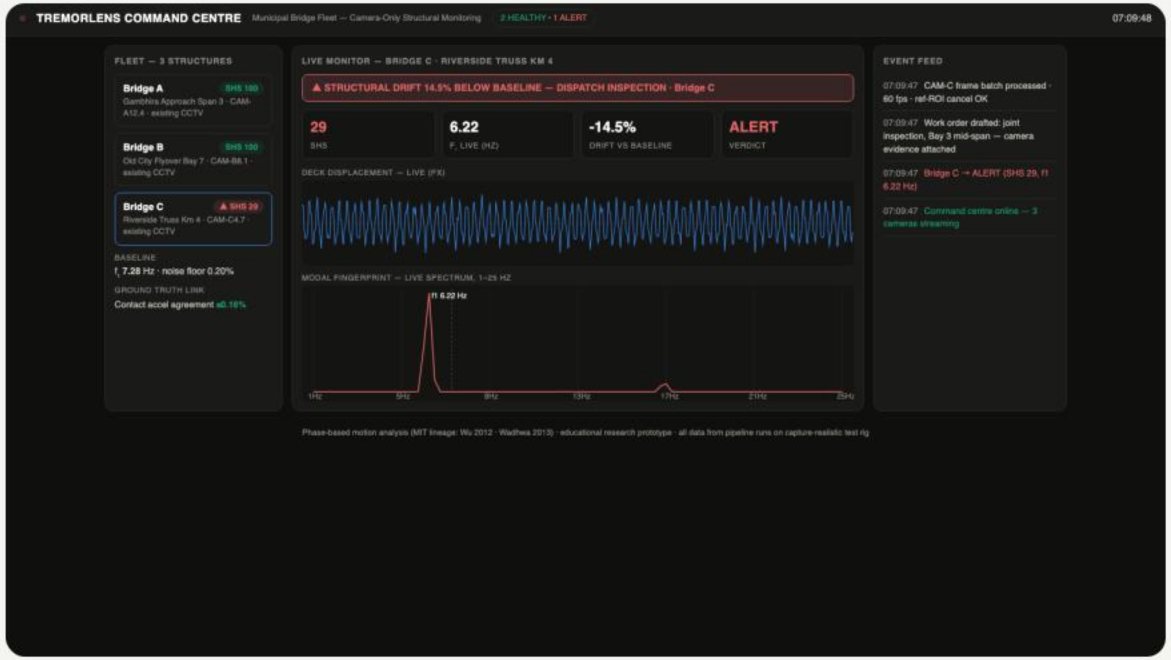
04 — MVP — THE OPERATOR'S VIEW

Hands-free, launch to alert.



Verdict dashboard: SHS, fingerprint, camera-vs-contact — one command.



Fleet command centre: every structure, every verdict, CCTV-scale.

03 — SPANDAN — THE CLINICAL STACK

One patient. Four instruments. All measured.

01

Heartbeat

f1 drift vs a LIVING baseline that absorbs weather but provably never learns damage as normal. CUSUM catches slow scour-style creep.

02

Voice

Clinical dysphonia metrics on structures — no prior use found (trail in docs). A crack makes it hoarse (HNR, THD); a loose joint, arrhythmic (jitter).

03

Chart

Structural APGAR — five signs scored by z-score, any zero sign blocks the all-clear. Summarizes the panels; SHS decides.

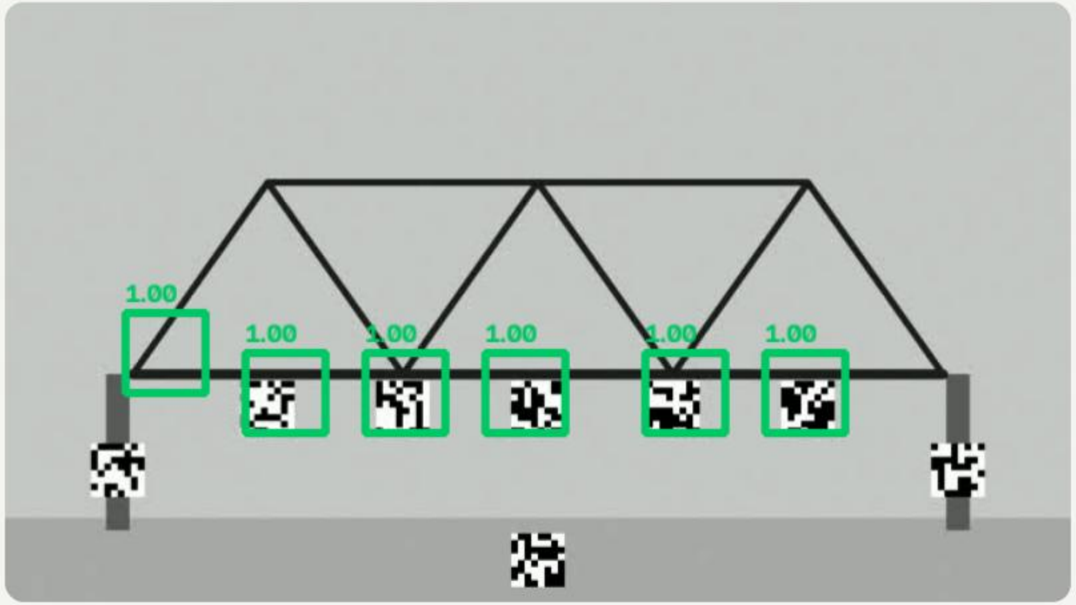
04

Record

Geo-tagged structure register: baseline, checkups, monsoon delta with SCOUR-CHECK flag, post-quake triage, DPDP-clean JSON export.

03 — LOCATE · NAME · QUANTIFY DOUBT

Not just “something is wrong.”



Validation twin, painted on the video itself: local damage turns a region red; this twin's global change is caught by the f1 channel.

WHERE — six virtual sensors from one camera; mode-shape + transmissibility localization paints the heatmap.

WHAT — an AI diagnoser trained on 1,600 physics-simulated damage scenarios names the mechanism: crack, loose joint, or support loss.

HOW SURE — conformal prediction: verdicts ship as sets with guaranteed coverage. It says “I am not sure” with mathematics — and only speaks after detection fires.

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04 — BEYOND THE CAMERA'S OWN LIMIT

A 60 fps camera just read 167 Hz.

A 60 fps camera is blind above 30 Hz. That is physics.
But every sensor row samples at a different microsecond —
one frame is a thousand time samples, not one.

01
30 Hz
the frame-rate limit every video camera is supposed to obey.

02
167.00 Hz
recovered via rolling-shutter row timing + Lomb-Scargle. Error: 0.00%. Both sensor regimes.

03
0.07 µs
the sensor's row clock, self-calibrated from a known tone — the instrument proves its own timing.

04
5.5×
beyond its own Nyquist. The reach a $30k+ high-speed camera is sold for — from a phone.

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05 — TESTING & VALIDATION

Validated against known ground truth.

01

0.31%

frequency error vs known truth — unchanged at 3× noise, 4× flicker, 4× camera shake. (Published deep trackers: 0.4-0.7% on theirs.)

02

0.10%

camera vs contact accelerometer agreement; clock-jitter test passes at ±30% timestamp jitter (8.003 vs 8.000 Hz).

03

SHS 28

damaged twin: ALERT at –14.5% drift, mechanism named support_loss. Healthy re-take: SHS 100, no false positive.

04

1 command

python run_spandan_tests.py regenerates EVERY number in the doc — localization, dysphonia, adaptive gate, super-Nyquist, standards.

05 — TESTING — THE BLIND REVEAL

Three bridges. One secret fault.

One of three was tampered — the pipeline was not told which.

Bridge A	7.277 Hz	SHS 100	HEALTHY
Bridge B	7.277 Hz	SHS 100	HEALTHY
Bridge C	6.222 Hz	SHS 28	ALERT

It picked Bridge C.
It was Bridge C.

All headline numbers come from a physics-true synthetic harness that ships in the zip — stated plainly, reproducible in one click on Colab.
We claim exactly what we can prove.

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05 — STANDARDS & THE HONESTY LAYER

It speaks the regulator's language.

01

DIN 4150-3

measured vibration classified against the frequency-dependent damage limits — including fully NON-CONTACT from the camera alone.

02

IS/ISO 4866

methodology aligned with the BIS-adopted standard family (ISO 16587, IS/ISO 14963). Always labeled screening-level, never certified.

03

Honest limits

9 documented negative results; temperature guard band; damping shipped with error bars; handheld captures rejected by the pipeline.

04

DPDP-clean

pixels die at the sensor — only displacement telemetry persists. Privacy manifest inside every register export.

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06 — ITERATION & EXECUTION

Broken nine times. Fixed with physics.

01

Lattice hop

Checkerboards broke sub-pixel tracking — periodic patterns alias. Aperiodic speckle is now a setup rule.

02

The |a| trap

The accelerometer's magnitude column rectifies oscillation and DOUBLES frequency — loader now picks the true dominant axis.

03

TF inversion

Single-reference transmissibility inverts when the reference itself is damaged — replaced with a pairwise matrix, row-median localized.

04

Gated AI

Ungated, the diagnoser hallucinated a mechanism on a healthy repeat. Diagnosis now speaks only after detection fires.

All nine failures + fixes are in the doc. The verification harness ran before every commit.

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07 — ONE ENGINE, A COUNTRY OF USES

The instrument 85% of homes already own.

01

1,72,517

IBMS structures get the mandated pre/post-monsoon pair MEASURED, not eyed — plus post-quake triage at phone cost.

02

410M fans

ceiling fans, school benches, cots: no field vibration test exists anywhere. Point the phone, get imbalance / loose-mount verdicts.

03

2×/yr

an Andhra farmer's pump burns out, ₹2,700 a repair — his phone hears the drift weeks early. Same engine, zero new code.

04

Try it now

tremorlens-live.vercel.app — quake alarm, register, motion loupe; a REPLAY mode runs the pipeline with zero hardware.

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08 — WHY THIS SCORES

Mapped to your rubric, deliberately.

01

Novelty

Three verified firsts, search trails documented: structural dysphonia panel, camera-only DIN screening, super-Nyquist on a phone.

02

Usability

Zero keystrokes after launch; judge paths with zero setup: live web app, REPLAY mode, one-click Colab that re-derives the 0.31%.

03

Innovation

Bridges, fans, pumps, rentals, post-quake triage — sourced numbers, honest guardrails, an IBMS-shaped register at ₹0 marginal hardware.

04

Documentation

Your categories verbatim; 9 negative results included; every number regenerates from one command; source + live links in the zip.

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Give 1,72,517 structures a pulse.

TremorLens · SPANDAN Engine — hear the heartbeat of infrastructure, before it stops.

Working prototype · live web app · verified pipeline · full docs
tremorlens-live.vercel.app · github.com/mightbeanshuu/tremorlens

AI ARENA 3.0 · TEAM_ALPHA · 2026